The Codon Lottery

A team game that turns RNA codons into amino acids — and amino acids into words

WHAT STUDENTS PRACTISE

- Reading the genetic code: RNA codon → amino acid.
- Using the one-letter amino-acid code.
- Seeing that a "message" can only be read correctly if you decode it correctly.

Materials

- One copy of the **codon chart** per team (below), or the full table from the Day 1 document.
- A set of **codon slips** per team (print and cut the sheet at the end of this document).
- A container to draw from (a bag or bowl), pencils, paper, a timer.

Procedure

1. **Prepare.** Print the slip sheet, cut it into 48 slips, fold them and place them in the container. Make one set per team.
2. **Hand out** a copy of the codon chart to every team (or student).
3. **Draw.** Each team draws **10 slips**. Each slip is an RNA codon. If a team draws a **stop codon** (UGA, UAG, UAA), it goes back into the container and the team draws again, so every team ends up with 10 amino-acid slips.
4. **Decode.** Use the chart to turn each codon into its amino acid and write down the **one-letter code**.
5. **Build words.** Rearrange the letters to make as many words as possible. Each letter slip can be used once per word, and a letter may be reused in different words. Words must have at least 3 letters.
6. **Score.** 1 point per letter of every valid word. Bonus: +5 for a word that uses all 10 letters, and +3 for a word on the science list in the answer key. The team with the highest score wins.

TEACHER TIP

Run the first round with the chart visible on the board and decode two codons together. For the second round, remove the chart from the board and let teams work from their own copy. Ask: "Why can two different codons give the same letter?" (The code is redundant — see the extension below.)

Codon chart used in this game

RNA codon	Amino acid	RNA codon	Amino acid
GCU	A (Alanine)	ACU	T (Threonine)
UGU	C (Cysteine)	UAU	Y (Tyrosine)
GAA	E (Glutamic acid)	UGA	Stop codon
AUU	I (Isoleucine)	UAG	Stop codon
AAG	K (Lysine)	UAA	Stop codon
CUU	L (Leucine)		
AUG	M (Methionine)		

Answer key and word bank

Only the letters A, C, E, I, K, L, M, T, Y can appear, so teams will not find every word. Some words they can make:

Length	Examples
3 letters	ACT, AIM, ALE, ATE, CAT, ICE, KEY, KIT, MAT, YAK, YET
4 letters	CAKE, CALM, CLAY, LAKE, LIKE, LIME, MAIL, MAKE, MILE, MILK, TAIL, TAKE, TALE, TALK, TICK, TILE, TIME
5–7 letters	TICKLE, TACKLE, CLIMATE (uses seven different letters — a superb find!)
Science words (+3)	CELL, CLIMATE, METAL, ACETYL

Note the slip set has few K, Y and C slips on purpose, so long words are rare and valuable.

Extension: redundancy of the code (level 2)

Give faster teams a second set with different codons for the same letters — for example GCC, GCA, GCG (all Ala = A), GAG (Glu = E), AUC, AUA (Ile = I), CUC, CUA, CUG, UUA, UUG (Leu = L), ACC, ACA, ACG (Thr = T). Ask them what they notice: several codons give the same amino acid. This is called degeneracy (redundancy) of the genetic code.

Second game: GeneDetectives

A short, 15–20 minute DNA-profiling mystery for teams of three. It introduces how forensic scientists compare DNA. The case below is fictional and complete, so you can run it straight away.

BACKGROUND: STRS
Forensic DNA profiles are based on short tandem repeats (STRs) — short DNA sequences repeated several times in a row at a fixed place (a locus) in the genome. People differ in the number of repeats. Everyone has two versions (alleles) at each locus, one from each parent. A profile is the list of repeat numbers at several loci.

The case

A laboratory has been broken into. A drop of blood was left at the scene. Four people were in the building. Their STR numbers (repeats per allele) are:

Person	Locus 1	Locus 2	Locus 3
Blood at scene	8, 12	15, 17	9, 11
Suspect A	8, 10	15, 17	9, 9
Suspect B	8, 12	15, 17	9, 11
Suspect C	12, 12	14, 17	9, 11
Suspect D	8, 12	15, 17	11, 11

Rounds

1. **Round 1 — Loci 1 and 2 only.** Which suspects can be ruled out? *(A and C do not match; B and D match at both loci.)*
2. **Round 2 — Add locus 3.** Who is left? *(Only B matches at all three loci. D is excluded at locus 3.)* Discuss: why was two loci not enough?
3. **Round 3 — Family connections.** A mother has alleles 10, 11 at locus 1 and a father has 8, 12. List every genotype their children could have *(10,8 · 10,12 · 11,8 · 11,12)*. Could a child with 8, 9 belong to this couple? *(No — 9 comes from neither parent.)*

Teams then present their reasoning. The most clearly argued conclusion wins the round.

A CAUTION FOR DISCUSSION
A DNA match is strong evidence, not magic. Real cases use 20 or more loci, and samples can be contaminated or mixed. Ask students what other evidence a detective would want.

Idea for older students: a digital game (concept)

GeneCraft — players act as genetic engineers in a futuristic lab, solving puzzles on DNA sequencing, CRISPR gene editing, genetic-circuit design and protein folding to save a fictional world from a virus. Levels get harder; points buy lab upgrades; a multiplayer mode allows teams to collaborate. Students could design the levels on paper as a project. (Game concepts contributed by Swarnendu.)

Printable slip sheet (48 slips)

Print one sheet per team and cut along the lines. Slip counts: GCU×7, GAA×8, AUU×7, ACU×6, CUU×5, AUG×4, UGU×3, AAG×3, UAU×2, and one each of the three stop codons.

GCU	**GCU**	**GCU**	**GCU**	**GCU**	**GCU**
GCU	**UGU**	**UGU**	**UGU**	**GAA**	**GAA**
GAA	**GAA**	**GAA**	**GAA**	**GAA**	**GAA**
AUU	**AUU**	**AUU**	**AUU**	**AUU**	**AUU**
AUU	**AAG**	**AAG**	**AAG**	**CUU**	**CUU**
CUU	**CUU**	**CUU**	**AUG**	**AUG**	**AUG**
AUG	**ACU**	**ACU**	**ACU**	**ACU**	**ACU**
ACU	**UAU**	**UAU**	**UGA**	**UAG**	**UAA**